Track C technical report - 15 October 2026

Track C: contrastive speech analysis

Technical report • 15 October 2026 • Controlled digital edits and a private natural-speech pilot

Problem and output

The system compares a reference reading and a same-transcript take, estimates word boundaries, extracts local acoustic measurements, marks candidate deviations on the take timeline, and displays a reference-relative operational score. It targets pauses, recorded-level drops, locally faster words and locally slower words. Candidate events include start/end times, measured cause and an edit-strength-like severity. They are not human-rated rhetorical quality judgments.

Data and provenance

Twenty-one archival speech excerpts from Reagan, Kennedy and Obama are candidate references. Each has 12 deliberately modified variants: four operators at three strengths. The 273 released WAVs include 252 digital variants. Audio is 16-kHz mono PCM16. Exact half-open edit intervals and source hashes are stored in the manifest and labels. All references are genuine archival speech crops; the altered versions are synthetic edits of those crops, never authentic mistakes by the speakers. RIGHTS.md lists per-source reuse evidence and limits. No Windows SAPI audio is included.

Train, validation and test are separated by speaker/source and whole passage family: Reagan 130 clips, Kennedy 39, Obama 104. Different eras, microphones and venues are confounded with split. The 22 original recordings from four local participants are private development material with no recorded redistribution consent and are excluded from the public package. Their instructions are not verified acoustic ground truth.

Alignment and acoustic features

Offline archival word alignment uses imported MFA TextGrids for all 273 clips. The 170 review flags mean disagreement with an older automatic aligner over 100 ms; they are not verified MFA errors. No human boundary accuracy is known. A live take uses PocketSphinx forced alignment when installed, otherwise a bounded MFCC dynamic-time-warping transfer of the reference word times. The fallback abstains for duration outliers, collapsed words, nonmonotonic mappings or high normalized path cost. It cannot verify that the words were actually spoken; transcript entry and an explicit listener confirmation gate that feedback.

Frames use a 40-ms window and 20-ms hop. Measurements include log-FFT energy, 13 MFCCs, estimated F0 and relative pitch in semitones, RMS level, spectral centroid and zero-crossing rate. Per-word durations and energy are compared at aligned word positions; interword gap excess measures pauses. Uncertain alignment and microphone gain limit interpretation.

Detector and scoring

Predicted candidate intervals

Inference takes only the two audio signals, their estimated word boundaries and an explicit

same-setup flag. It never reads a clip ID, flaw label, intended take or programmed edit interval. For word i, the duration ratio is take duration/reference duration; the level difference is 10 log10(mean take frame power/mean reference frame power). For adjacent words, pause excess is take interword gap minus nonnegative reference gap. Threshold crossings generate word or gap intervals, and same-type neighbors within 600 ms merge.

Event | Selected threshold | Cause shown in dashboard

Pause | Gap excess >=0.25 s | Seconds longer than reference gap

Quiet local level | Clip-relative difference <-6 dB | Within-clip word energy change

Rush | Tempo-normalized duration ratio <0.82 | Locally shorter word duration

Drag | Tempo-normalized duration ratio >1.30 | Locally longer word duration

Event merge | Same type separated by <=0.60 s | Coalesces fragments of one edited region

The duration ratio is divided by the pair’s median word-duration ratio, and word level change is centered by each clip’s median energy. This reduces uniform speaker-rate and gain differences while retaining localized deviations. Uniform whole-clip changes are shown separately. The optimizer precomputes candidate word/gap evidence, tests 1,024 threshold/merge combinations on Reagan training families, ranks by mean family F1, and uses Kennedy validation only as a guard against lower precision or F1. The Obama test speaker is not loaded during selection. The 0.60 s merge setting may combine separate natural events and requires review on natural speech.

Transparent rubric

Three 0-10 operational dimensions cover pacing, pauses and local level. Merged events are scored by unique affected words with burden 1, 2 or 3 for minor, moderate or severe; this avoids multiplying a single displayed span by its number of word events. Pause score falls by 3 points per cumulative second of excess pause. The combined score is the mean of the three available dimensions. It is not a human preference score, speaking skill rating or MOS. Transcript or alignment failure withholds word-level measurements. Appropriate rhetorical pauses may be flagged.

Dashboard

The interactive evaluation.html page overlays reference and take curves by transcript word position, highlights predicted regions in orange and archive edit intervals separately in blue, and lists each explanation. It accepts a take and transcript for a matching archive passage. The public pair.html page accepts two arbitrary recordings for coarse activity, pause and level comparison; it does not claim word-level localization. Both pages use the local server. The human.html page and private pilot audio are excluded from the public package.

Evaluation, limits and reproduction

Held-out digital-edit results

One ground-truth edit exists per non-reference clip. A predicted event counts as localized

only when its type matches and its interval IoU with the exact edit is at least 0.5. Event precision uses all predicted intervals; recall uses all labeled edits. This makes extra detections visible.

Split | Labeled edits | Type present | IoU>=0.5 recall | Event precision | Event F1

Train / Reagan | 120 | 71.7% | 58.3% | 52.2% | 55.1%

Validation / Kennedy | 36 | 75.0% | 55.6% | 30.3% | 39.2%

Test / Obama | 96 | 75.0% | 60.4% | 42.3% | 49.8%

On the test speaker, pause localization reached 23/24, quiet 9/24, rush 14/24 and drag 12/24. By severity, minor recall was 7/32 (21.9%), moderate 23/32 (71.9%), and severe 28/32 (87.5%). At IoU >=0.3 recall was 67.7%; with a –200 ms onset/offset collar, recall was 47.9% and precision 33.6%. Family-bootstrap 95% intervals at IoU >=0.5 were 49.0-67.7% recall, 31.1-57.4% precision, and 39.1-61.1% F1. There were 79 extra or insufficient-overlap predictions, down from 106 with the previous detector. Mean onset/offset errors among IoU-localized events were 0.072/0.166 s.

One-edit benchmark type gate

For the benchmark’s known setup of exactly one programmed edit per clip, we tested a simple inference-time filter: the paired classifier predicts the edit type, then the localizer retains only candidate intervals of that type; if the classifier predicts no edit, it suppresses candidates. The filter threshold was chosen using five-fold passage-family grouped out-of-fold predictions on Reagan training data and a Kennedy validation guard, without loading Obama test outcomes for selection. On the held-out Obama test split, precision increased from 42.3% to 65.2% and event F1 from 49.8% to 62.7%, while recall remained 60.4%; candidate intervals fell from 137 to 89 and the 58 correctly localized edits were retained. Family-bootstrap 95% F1 interval was 50.5-73.3%. This is a useful prototype result, but the same fixed test split had already been used for earlier classifier reporting, so it is not a fresh independent benchmark. Because natural speech may contain multiple event types, this gate is benchmark-only and is not applied to user uploads.

Retrained edit-signature classifier

To improve the separate whole-clip task, logistic regression, RBF SVM, Random Forest, ExtraTrees and histogram gradient boosting were compared with five-fold grouped cross-validation over Reagan training passage families. Kennedy validation guarded against regression; it and the training families were then used to refit the selected ExtraTrees heads. The frozen model scored 104/104 (100.0%, macro F1 1.000) for digital edit type and 101/104 (97.1%, macro F1 0.977) for programmed edit strength on the Obama test clips. This is the same fixed test set previously reported for the logistic baseline (97.1% type accuracy; 73.1% strength accuracy), so it is not a new independent benchmark. The candidate search did not use test outcomes. These are synthetic-operation recognition metrics, not speech-quality or natural-speech scores. The current model and protocol are documented in MODEL_CARD.md.

Limits and remaining human validation

The detector has low precision, especially on Kennedy, and word-alignment errors can create

misleading regions. Quiet events are only recorded-energy differences; automatic gain control and microphone changes can reverse their meaning. The same-reference archival variants are far easier than unrelated natural performances. There is no verified natural-speech accuracy, human severity calibration, independent transcript audit or measured word-boundary error. Do not present this as a reliable judge of people or public speakers.

A nuisance negative-control run compared all 21 unedited reference clips with global –10% speed, –6 dB gain, –3-semitone pitch, stationary noise at 25 dB SNR, and 300-3400 Hz band-limited versions. It produced zero localizer events in each of the eight controls. This checks false alerts under global changes on these same-source pairs; it does not prove cross-speaker performance or verify that local flaws remain detectable after those changes. MP3 encoding was not tested.

Reproduction

Install requirements.txt, run python src/build.py and python src/validate.py; use the supplied MFA alignments; run python -m unittest discover -s src -p 'test_*.py', python src/tune_localizer.py, python src/evaluate_localizer.py, python src/make_viewer_data.py, and python src/serve.py. Open http://127.0.0.1:8765/evaluation.html or /pair.html. Arbitrary pair comparison reports coarse speech activity, pauses and level; word-level localization is limited to aligned same-transcript pairs. Dataset hashes, edit labels, split manifests, event predictions, thresholds, test metrics and code are included. Unit tests and structural validation are separate from perceptual validation.

References: official Track C problem statement and submission guide supplied for the hackathon; MFA documentation; DTW implementation and source/rights links recorded in RIGHTS.md. The combined benchmark result is specific to controlled digital edits and does not establish accuracy on natural speech. The user handles the required submission video separately.